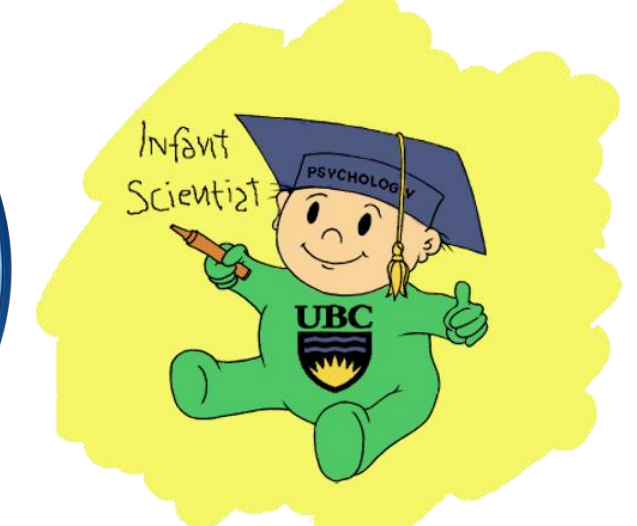




Developmental changes in infants' sensitivity to vowel harmony for word segmentation

Helen Shiyang Lu^a, Alexis K. Black^a, & Janet F. Werker^b^a School of Audiology and Speech Sciences and ^b Department of Psychology, University of British Columbia

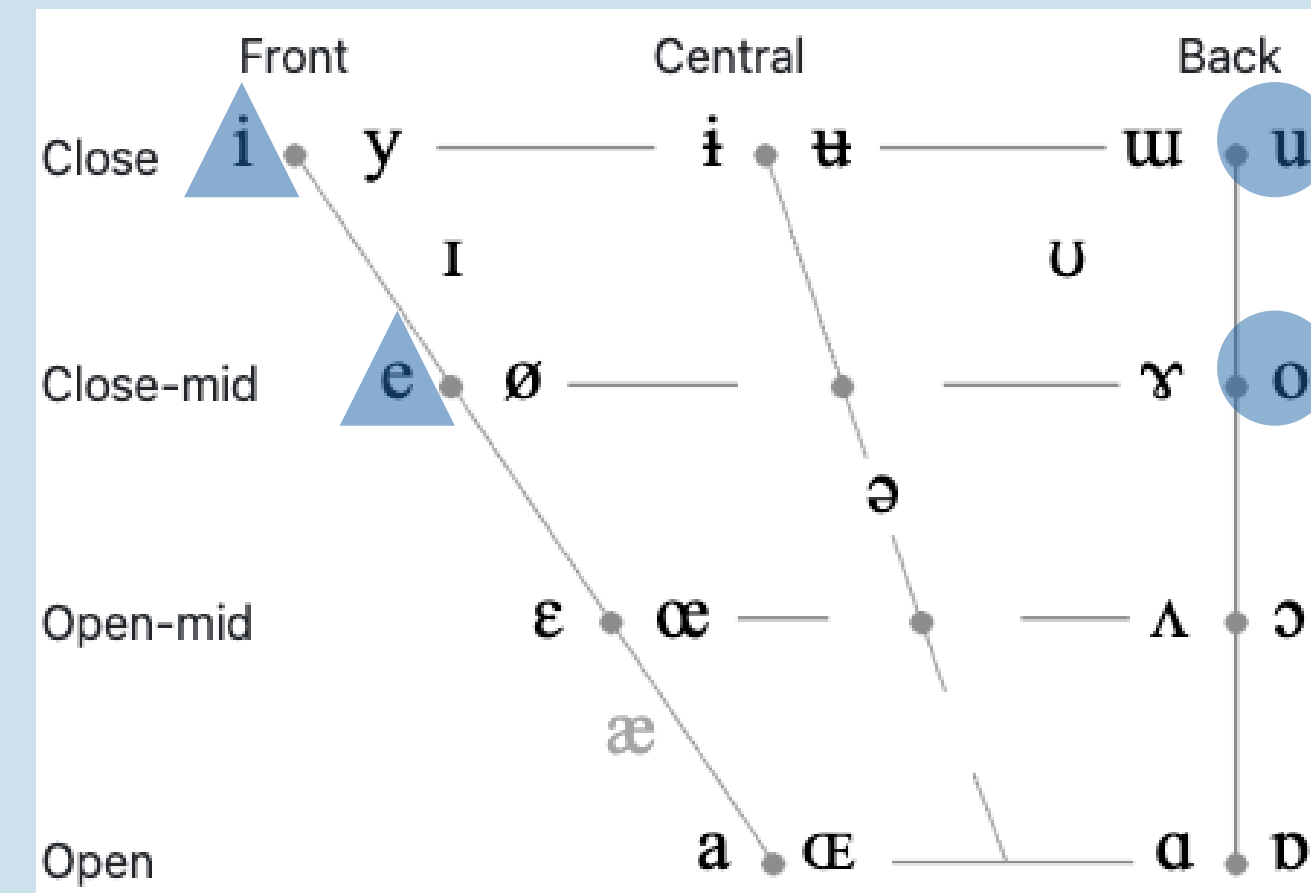
Scan to download



Contact: helens.lu@ubc.ca

Introduction

- Vowel harmony (VH)** is a phonological process in which vowels within a word agree on certain phonetic feature¹
 - e.g., backness in Turkish *çocuk-lar* “children” *ev-ler* “houses”
- Disharmonies can signal word boundaries, making VH a potential cue for word segmentation²
- Infants learning VH languages (e.g., Turkish³ and Akan⁴) distinguish harmonic vs. disharmonic words by 6 months, and use VH for word segmentation by 9-11 months⁵
- Recent evidence suggests an **early, broad sensitivity to VH**:
 - 4-month-old infants primarily exposed to non-VH languages prefer harmonic compared to disharmonic sequences⁶
 - 7-month-old English-learning infants (a non-VH language) can segment speech using VH cues⁷
- This sensitivity **attenuates without relevant input**:
 - By 7-8 months, infants exposed primarily to non-VH language(s) no longer prefer harmonic words^{6,7}, a pattern consistent with perceptual attunement
- Bilingual infants may follow a different developmental path⁸:
 - Experience with multiple languages could maintain broader perceptual sensitivity⁹
 - However, exposure to more than one non-VH languages may also accelerate the down-weighting of VH as a cue



Methods – Central Fixation

Baseline Phase (testing preference)

- 4 fixed-length trials with a disyllabic word repeated 16 times, with 500-ms interstimulus interval (ISI)

ke-mi

vs.

*ki-mo**no-gu**nu-ge*

Familiarization Phase

- 108-s continuous speech stream (45 repetitions) with no acoustic, prosodic, or statistical cues to word boundaries

de-bi-do-tu-be-pi-to-pu

Test Phase (testing segmentation)

- 8 infant-controlled trials presenting a disyllabic sequence from the familiarization stream with 500-ms ISI

to-pu

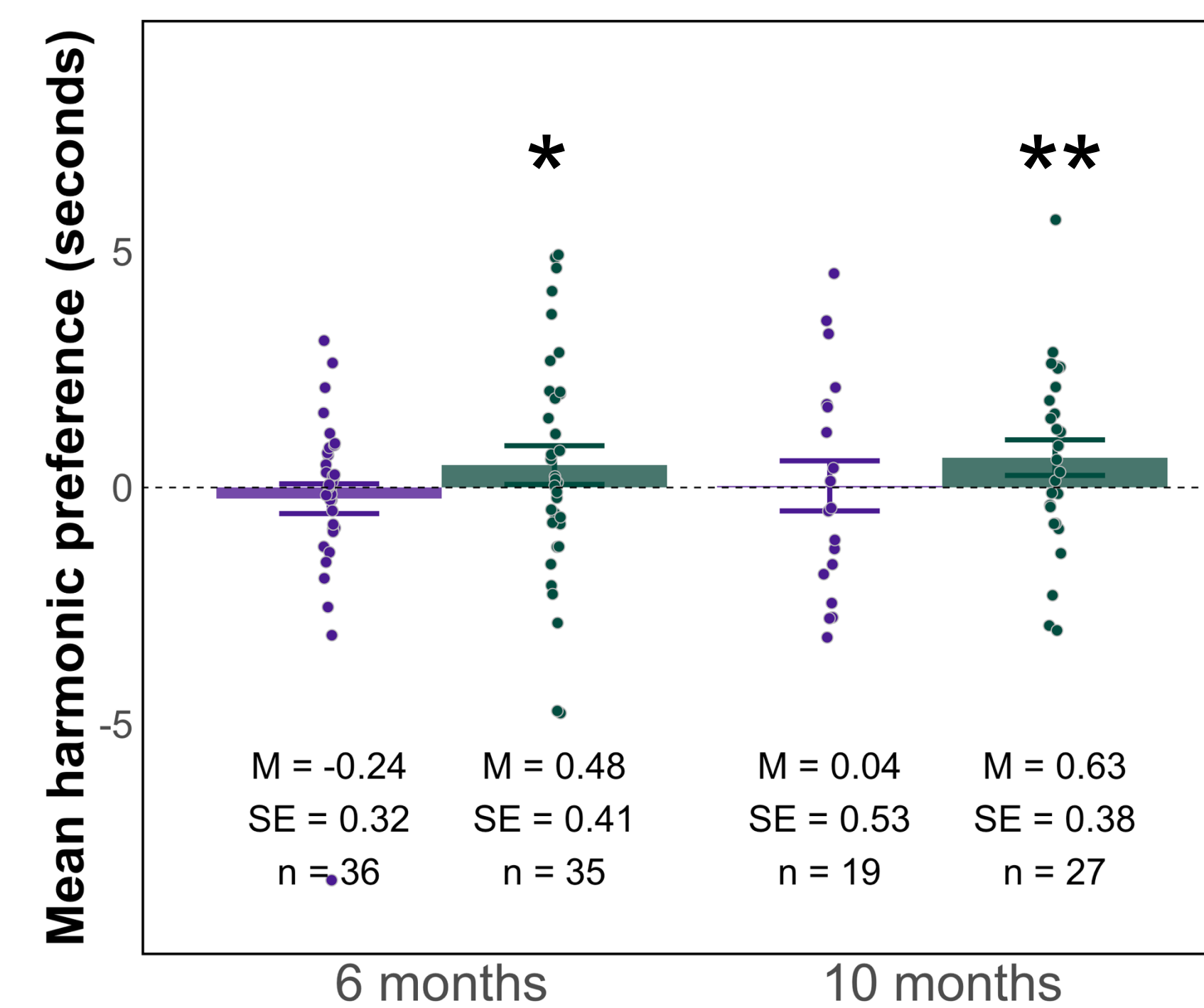
vs.

*bi-do**de-bi**tu-be*

Preliminary Results

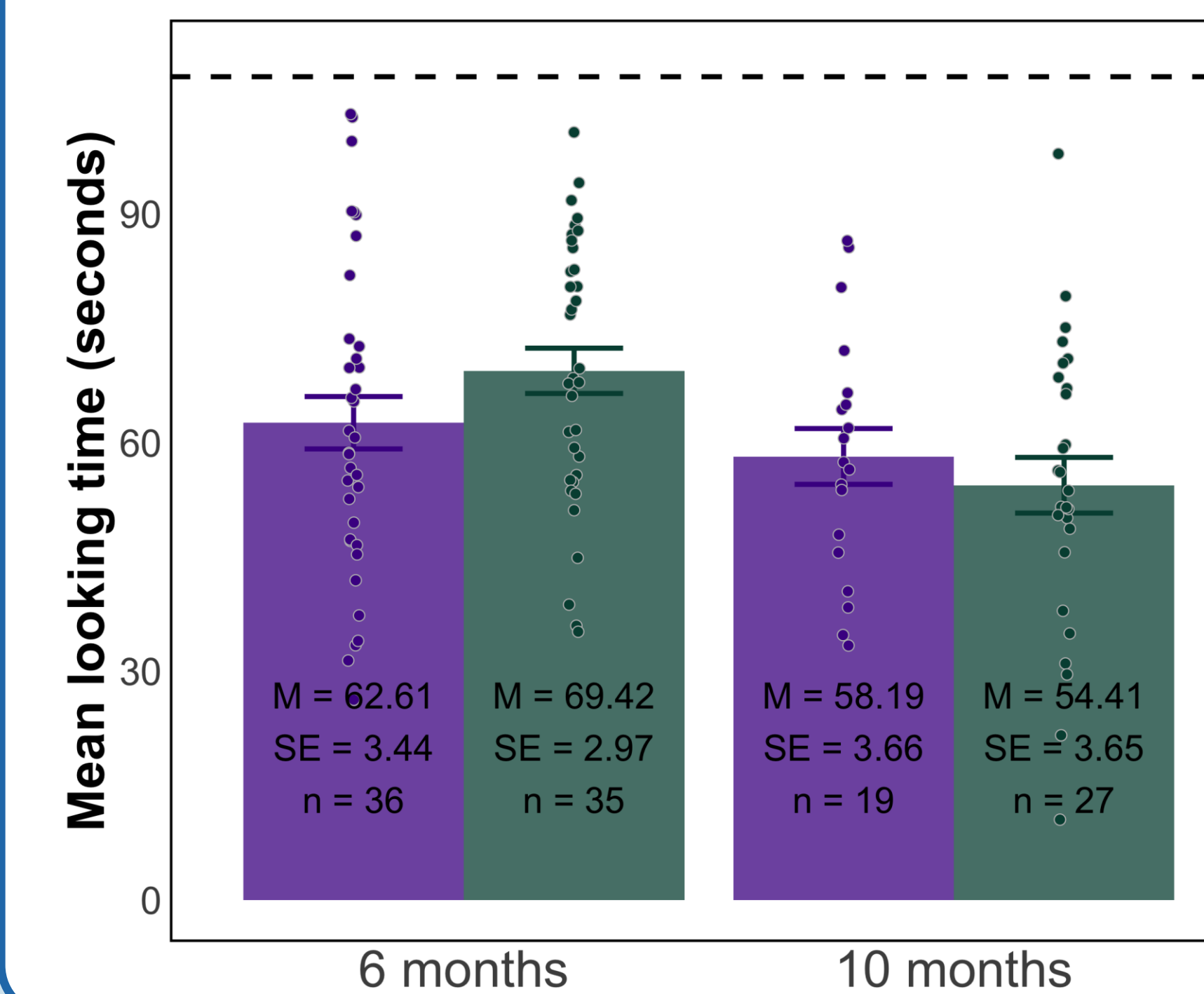
At baseline, infants in general showed a **harmonic** preference ($p = .046$), and this preference was more pronounced in the **monolingual** infants ($p = .018$).

■ Bilinguals ■ Monolinguals



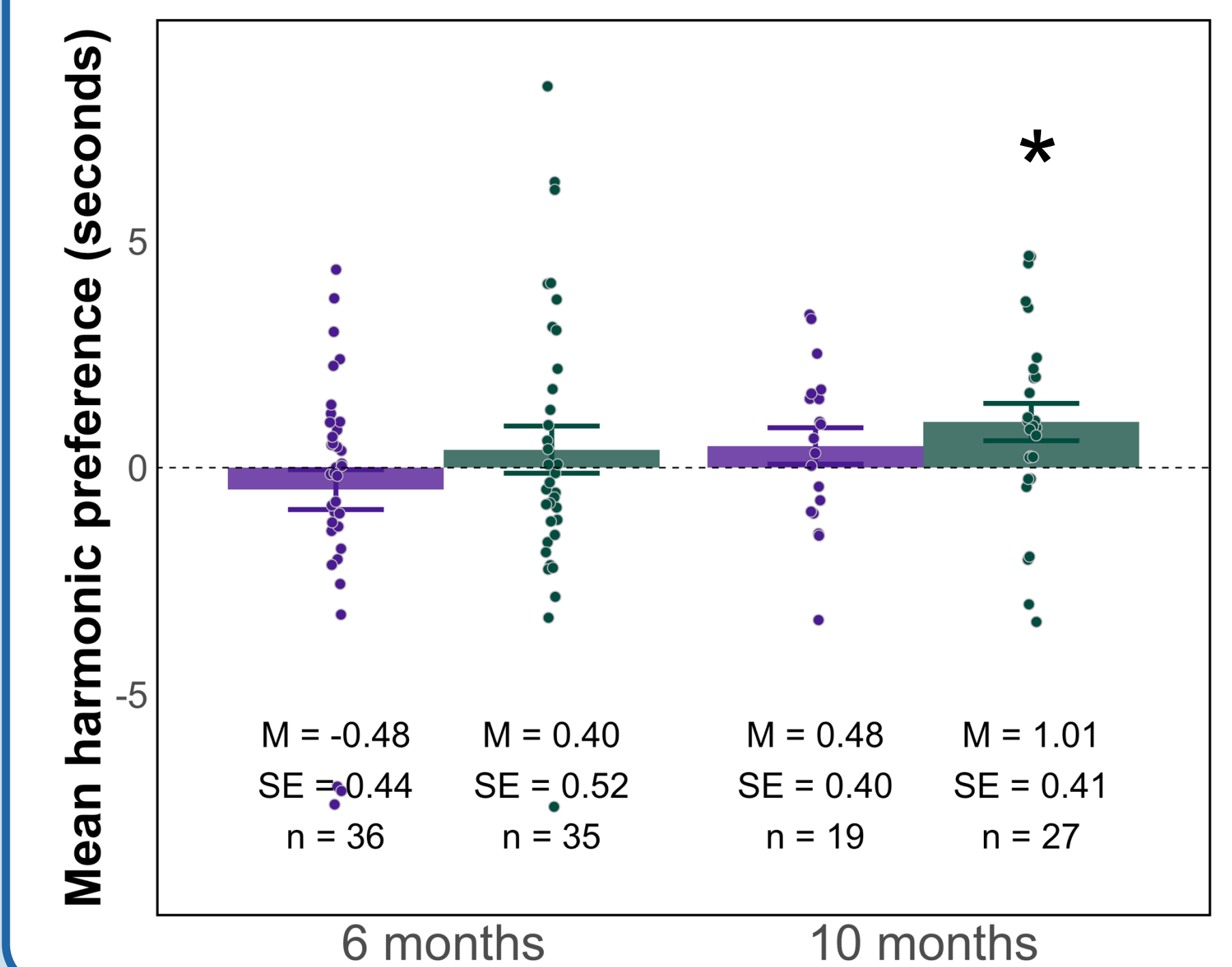
In familiarization, 10-month-olds looked shorter than the 6-month-olds ($p = .007$), and no significant interaction was observed for age x bilingualism.

■ Bilinguals ■ Monolinguals



At test, we did not find evidence for successful segmentation overall, but 10-month-old **monolinguals** showed a harmonic preference ($p = .043$).

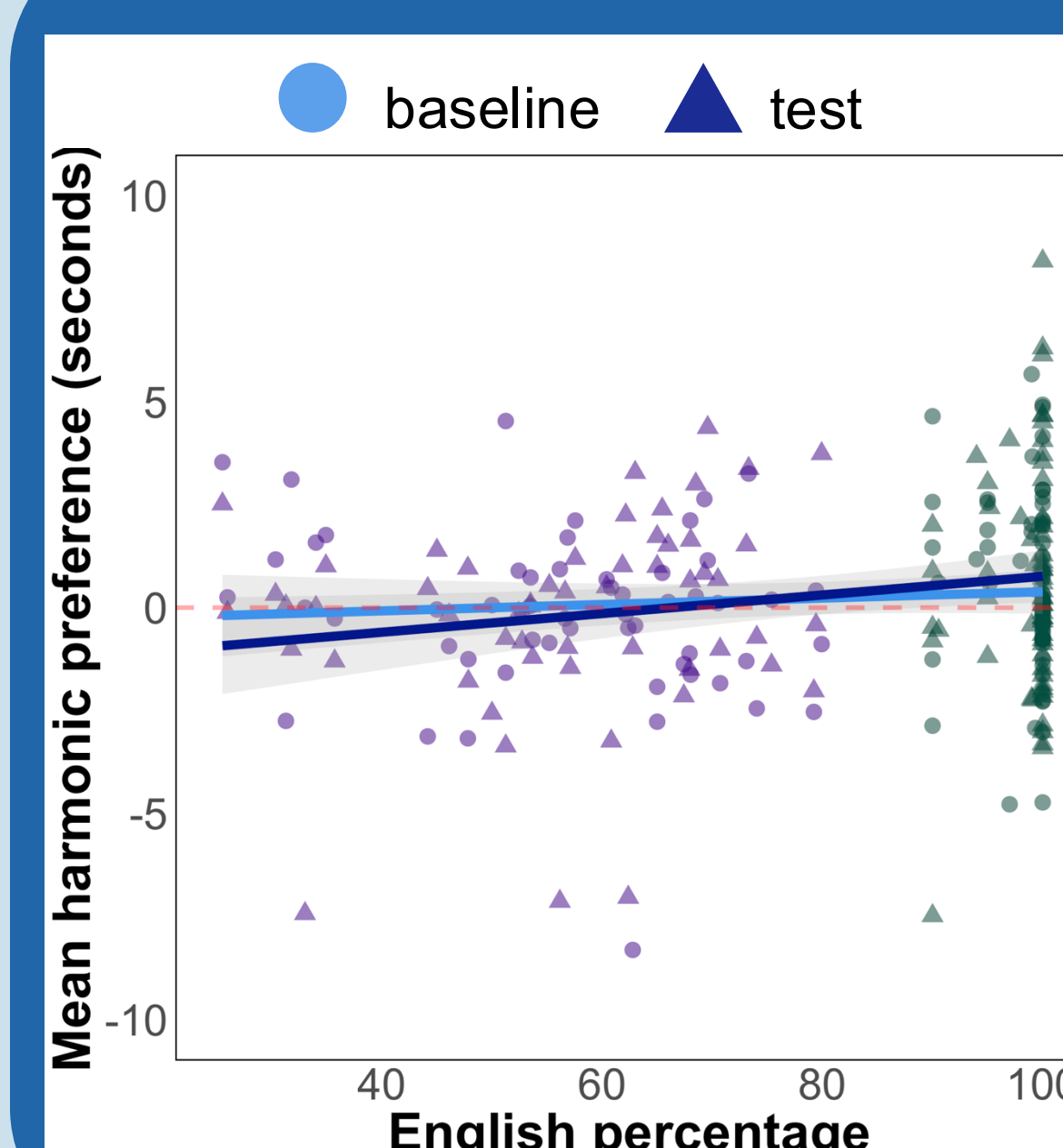
■ Bilinguals ■ Monolinguals



Does VH sensitivity decline in infants with no exposure to VH, and does exposure to more than one non-VH language change this trajectory?

Participants

- 35 **mono-** and 36 **bilingual** 6-month-old infants
- 27 **mono-** and 19 **bilingual** 10-month-old infants
- All infants had no to minimum (< 5%) exposure to VH



Summary & Discussion

- English monolinguals preferred harmonic words at baseline:
 - Adding English exposure yielded an interaction with harmony at baseline ($p = .093$) and at test ($p = .047^*$)
 - Preliminary analysis of infant-directed corpus¹⁰: ~70% English disyllabic words agree in rounding
- No clear evidence for infants using VH for word segmentation: Persistent preference across phases in 10-month-old monolinguals obscures segmentation effects
- Bilingual exposure**: Lack of VH sensitivity in bilingual infants suggest either (1) English input drives the sensitivity, or (2) bilingual exposure accelerates the decline